



Swinburne University of Technology Hawthorn Campus Department of Computing Technologies

COS20028 Big Data Architecture and Application Lab Task 2 - Semester 2, 2024

Assessment Title: Write a MapReduce Program for Finding the Line Count in the input source.

Assessment Outcome: Pass/Fail (for practice)

Assessable Item:

- One (1) piece of a report containing the complete codes of the driver and mapper with the screenshot of submitted commands and execution results.

Purpose of Assignment

This assignment evaluates whether a student can utilise MapReduce flexibly and knowing where to make corresponding changes for implementing a customised function.

This is an individual assignment and is mandatory for passing this unit.

Introduction

The source data is built in the virtual machine image under the path of **/home/training/training_materials/developer/data**. Our goal is using MapReduce to find out how many lines are included in the **"bible.tar.gz"** file.

Assignment Goal

This assignment aims at training students to get familiar with the major components in MapReduce, recognising the functionalities of different components, and identifying the right place for implementing functions to accommodate the requirements.

Assignment Task

Write a MapReduce program to find the number of lines included in the input source. You are asked to use the built-in reducer class called **"IntSumReducer"** but not building your own reducer. To use the system built-in **"IntSumReducer.class"**, you need to import **"org.apache.hadoop.mapreduce.lib.reduce.IntSumReader"** and then call it in your driver code. You must use **System.out.print()** or **System.out.println()** to print your name and student ID on the screen while executing from the driver code. Please use the template to prepare your submission.

Submission Requirement

Only the submission using the given template will be evaluated by the tutors.

Failure to adhere to the submission requirements will immediately result in a failure for this assignment.

Rubric: Lab Task 2

Task	Assessment Outcome
Successfully finding the answer to the given question in the template and the submitted code in the report is reasonable, complete, and fully functional.	Pass/Fail

----- End of Assignment -----